

Question Number	Answer	Additional guidance	Mark
8(a)	An answer that makes reference to the following points: • correct example of molecular evidence (1) • to see if the sequences from Akhal-Teke and Turkoman breeds are {the same / very similar / few differences} (1)	e.g. {DNA / RNA / gene / allele / amino acid} sequences ignore genotype ignore {molecular phylogeny / DNA analysis} unqualified ignore breeding together to form fertile offspring ignore {similar / same} {genes / genotype}	(2)

Question Number	Answer	Mark
8(b)	The only correct answer is B 3607 <i>A is not correct because $0.661 \times 5457 = 3607$ (to whole number)</i> <i>C is not correct because $0.661 \times 5457 = 3607$ (to whole number)</i> <i>D is not correct because $0.661 \times 5457 = 3607$ (to whole number)</i>	(1)

Question Number	Answer	Additional guidance	Mark
8(c)(i)	An answer that makes reference to the following point: larger hooves {prevent the Arabian from sinking into the sand / allow walking in the sand} (1)	ACCEPT large surface area (of hoof) {helps movement on the sand / prevents sinking} ignore they live in different {areas / habitats}	(1)

Question Number	Answer	Additional guidance				Mark												
8(c)(ii)	An answer that includes the following points: • p correctly calculated (1) • q correctly calculated (1) • correct calculation of number of horses with heterozygous and homozygous recessive genotype which add up to 354 (1)	<table><tr><td>genotype</td><td>two copies of the dominant allele</td><td>heterozygous</td><td>two copies of the recessive allele</td></tr><tr><td>Part of Hardy-Weinberg equation</td><td>p²</td><td>2pq</td><td>q²</td></tr><tr><td>Number of individuals</td><td>46</td><td>179</td><td>175</td></tr></table> ecf applies for mp2 only correct answers score full marks				genotype	two copies of the dominant allele	heterozygous	two copies of the recessive allele	Part of Hardy-Weinberg equation	p ²	2pq	q ²	Number of individuals	46	179	175	(3)
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*8(d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • there are more sperm that can swim in the fresh sample / converse for frozen and defrosted sample • sperm cells survive for longer in fresh sperm sample / converse for frozen and defrosted sample • fewer sperm cells have damaged DNA in fresh sperm sample / converse for frozen and defrosted sample • quantitative comparison, e.g. double in fresh, 59% decrease in survival • as the percentage of sperm that can swim increases then the number of successful pregnancies increase / positive correlation • the closer the correlation coefficient number is to +1.0 then the stronger the correlation between the two variables • correlation coefficient is +0.90, which is a strong positive correlation advantages of using frozen sperm samples in captive breeding programmes • can collect samples from {every male / lots of males / both captive and wild} • can do {breeding programmes / artificial insemination / IVF} • can increase population numbers (of the endangered breeds) / species does not become extinct • do not need to {transport / find} {wild / male} horses / can use samples from horses that have died / if all males died out the females could still be impregnated • idea that breeders can select which sperm sample would {maintain genetic diversity in the population / prevent decrease in genetic diversity / prevent inbreeding} • frozen sperm samples {take up less space than a male horse / can be stored for a long time} disadvantages of using frozen sperm samples • reduced chance of successful {fertilization / pregnancy} in females (using frozen sperm samples) because {reduced ability of sperm to swim / sperm may not reach egg cell to fertilise it / reduced sperm survival / reduced speed of swimming} • {damaged sperm DNA / mutations} will mean {fertilisation will not occur successfully / fewer zygotes / embryos don't develop / miscarriages occur / offspring will have genetic diseases} • cost of {equipment / procedure} / specialist staff required / storage space required • not suitable without females (of reproductive age) / reduced chance of {fertilization / pregnancy} could result in breed becoming extinct

		Additional guidance
Level 0	0	No awardable content
Level 1	1-2	Demonstrates isolated elements of biological knowledge related to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of biological facts / concepts. Consequences are discussed which are supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.